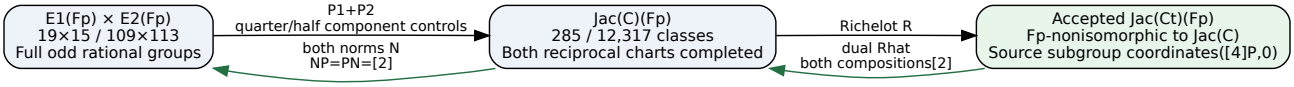
Explicit elliptic coordinates for the rational groups

2026-11-11 · Chained ECDLP Transfers · CT17

Unsigned/unverified standing. All285 F17 and12,317 shiftedF101 classes pass both quotient norms,reciprocal completion and odd-component reconstruction. Accepted target source points have coordinates([4]P,0). PCC036d81/96b1e2 bind correctness evidence,no DLP solver or measured gain. [pcc:unverified]



Complete reciprocal chart

For C:$y^2=x^6+a2*x^4+a4*x^2+b$,$b=s^2$ nonzero,set $T=s/x$,$Y=b*y/x^3$. The second monic cover is $Y^2=T^6+a4*T^4+a2*b*T^2+b^2$,with inverse $x=s/T$,$y=s*Y/T^3$. The two elliptic quotient cubics are [b,a4,a2,1] and[b^2,a2*b,a4,1]. No twist is silently substituted. [pcc:unverified]

The actual balanced divisor map transforms whole finite support. Degree lost at x=0 maps to the correct target infinity,including multiplicity2. Old infinities map to($T=0$,$Y=\pm b$); their hyperelliptic fiber carries K to the new K. The completed class retains all finite/infinity coefficients before reducing. Every rational class,both reciprocal inverses and paired additions pass; zero-support multiplicities0/1/2 are all covered. [pcc:unverified]

Full rational-group coordinates

The involutions sigma:$(x,y)\to(-x,y)$ and iota*sigma define the two quotients. Hyperelliptic iota induces-1 on Jac(C),so pull/norm operators are$1+$sigma and$1-$sigma. Thus their sum is[2],each norm/pull is[2],and cross-norms are0. This geometric bridge remains a written primary input. [pcc:unverified]

All285/12,317 classes satisfy these identities and reconstruction by halving each elliptic component. Independently counted component orders are19/15 and109/113. Norm pairs are distinct and exhaust the complete product. These are rational-group isomorphisms,not an assertion that the abelian varieties are isomorphic to products. Incorrect odd annihilators and unsupported scales/constants are rejected. [pcc:unverified]

Target coordinates and direct source controls

Let N=(N1,N2),P=P1+P2. With accepted Richelot R and dual Rhat, M=N*Rhat and S=R*P satisfy MS=SM=[4]. Their odd rational groups therefore have an explicit inverse by quartering elliptic components. This follows from the complete individual pull/norm/dual identities. [pcc:unverified]

For every original r19/r109 source scalar,coordinates are([4]P,0). Both components,zero second factor,source inverse5/82 and reversed-class reconstruction pass. No map reads a DLP scalar. The other elliptic factor supplies no extra DLP equation for this subgroup. Generic algorithms still face order r;a larger ambient Jacobian alone gives no generic-query gain. A non-generic solver would need separate complete cost evidence. [pcc:unverified]

Compiled algebra and exact receipts

Seven SplitQuotients.lean lemmas pass gate7,PCCfffa35: cleared reciprocal curve/inverse,complementary norm sum,odd-annihilator halving,conditional split reconstruction and target first/other components. Two broad tactic failures remain preserved; explicit local power identities give the pass. No new axioms,no finite-field/Picard geometry or group orders formalized. Total62 written project lemmas. [pcc:unverified]

Primary inputs remain Galbraith corrected ch.8§8.3/ch.10 and the existing bielliptic cover/pull-push family. No prior-art novelty claimed; KB/ecdb unavailable. Group arithmetic and harnesses remain producer-authored Cantor checks,without signed independent whole-map standing. [pcc:unverified]

New targeted full decomposition,source-component controls,formatting and strict Clippy pass. Current full49-test CI pending. Prior63cf80560 exact-head focused43024621372 passes native25/reference47,one old benchmark ignored, formatting/Clippy/guards(PCCbfa6ff); its exact scope is retained. [pcc:unverified]

Receipts: split-quotients-proposal-01,split-quotients-01, split-target-components-01,split-quotients-clippy-01. Source: src/balanced.rs,src/nonsplit_map.rs. Run consumer check.sh with Rust1.99.1 and unchanged crypto75a737b8 guards,or select the two new correctness tests. Run upstream proof-gate.sh on SplitQuotients.lean. Full account: SPLIT_QUOTIENTS.md. [pcc:unverified]

Next route and cost boundary

The rational Richelot next neighbors are already closed by forced dual backtracks. Next assess rational odd-prime maximal-isotropic kernels without pre-accepting a map. Retain original-source/direct elliptic controls for any candidate. Production/catalog/large fields stay open.

No benchmark,solver comparison,paid allocation or computational advantage was demonstrated. All future comparisons must include construction,transport, solver,recovery and verification. S3/live-budget,KB/vault and crypto frozen ledger/archive blockers remain; no cross-prime descent or cost edge promoted. [pcc:unverified]